Lecture Scribe Probability

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Below is a faithful, exam-ready **lecture scribe for Lecture 4 (part covered in this PDF)** strictly following the logical order and content of the lecture slides, without adding extra motivation or outside material.
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CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

(Portion Covered: Introduction to Probability Theory — Experiments, Sample Space, Axioms, Corollaries, Propositions)

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1. Introduction to Probability Theory

Experiments, Sample Space, and Events

Definition 1: Experiment (E)

An **experiment** is a procedure we perform that produces some result.

Example:
Tossing a coin five times is an experiment denoted by:

$$E_5$$

Definition 2: Outcome (ξ)

An **outcome** is a possible result of an experiment.

Example:
One outcome of the experiment E_5 (five coin tosses) is:

$$\xi_1 = HHTHT$$

Definition 3: Event

An **event** is a certain set of outcomes of an experiment.

Example:
Let C be an event in experiment E_5 defined as:

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

Definition 4: Sample Space (S)

The **sample space** S is the collection (set) of all possible distinct outcomes of an experiment.

Properties:

- Outcomes are **mutually exclusive**
  → Only one outcome can occur at a time
- Outcomes are **collectively exhaustive**
  → No outcome outside the set is possible

Example (single coin flip):

$$S = \{H, T\}$$

- Mutually exclusive: cannot get H and T together
- Collectively exhaustive: no other result possible

Continued Properties of Sample Space

- S is the **universal set of outcomes**

- Sample space can be:

1. Discrete
2. Countably infinite
3. Continuous

Examples of Experiments and Their Sample Spaces

1. Flipping a fair coin once
2. Rolling a cubical die with numbered faces
3. Rolling two dice
4. Flipping a coin until tails occurs
5. Random number generator in interval $[0, 1)$

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2. Axioms of Probability

Before assigning probabilities, probability is defined as:

> A numerical measure of the likelihood of events
> OR
> A function that maps events to numbers representing likelihood

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Axiom 1: Non-negativity and Boundedness

For any event A:

$$0 \leqslant Pr(A) \leqslant 1$$

Axiom 2: Probability of Sample Space

If S is the sample space:

$$Pr(S) = 1$$

Axiom 3: Additivity for Mutually Exclusive Events

If:

$$A \cap B = \emptyset$$

then:

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

Axiom 3 (Infinite Version)

For infinitely many mutually exclusive events $A_1, A_2, \ldots$ such that:

$$A_i \cap A_j = \emptyset \quad \text{for } i \neq j$$

then:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

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3. Corollary from Probability Axioms

Define $A_i = \emptyset$ for all $i > M$.

Corollary 2.1 (Finite Additivity)

For a finite number M of mutually exclusive events:

$$A_i \cap A_j = \emptyset \quad (i \neq j)$$

then:

$$Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} Pr(A_i)$$

Observations

- Axiom 3 reduces to Corollary 2.1 when the sample space is finite
- Axiom 3 is necessary for infinite sample spaces

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4. Propositions from Probability Axioms

Proposition 2.1 (Complement Rule)

For any event A:

$$Pr(A^c) = 1 - Pr(A)$$

Proposition 2.2 (Monotonicity)

If:

$$A \subset B$$

then:

$$Pr(A) \leqslant Pr(B)$$

End of Lecture Content Covered in This PDF

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All definitions, axioms, corollaries, propositions, and examples are written step-by-step exactly as presented in the lecture slides

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CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

(Covered Portion: Introduction to Probability Theory)

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1. Experiments, Sample Space, and Events

Definition: Experiment (E)

An **experiment** is a procedure we perform that produces some result.

Example:
Tossing a coin five times is an experiment denoted as:

$$E_5$$

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An **event** is a certain set of outcomes of an experiment.

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Definition: Sample Space (S)

The **sample space** S is the collection or set of all possible distinct outcomes of an experiment.

Properties:

• Outcomes are **mutually exclusive**
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• Outcomes are **collectively exhaustive**
→ No outcome outside the set is possible

Example (single coin flip):

$$S = \{H, T\}$$

- Cannot get H and T simultaneously (mutually exclusive)
- No other result exists (collectively exhaustive)

Continued Properties of Sample Space

- S is the **universal set of outcomes**
- Sample space may be:

1. Discrete
2. Countably infinite
3. Continuous

Examples of Experiments

1. Flipping a fair coin once

2. Rolling a cubical die with numbered faces
3. Rolling two dice
4. Flipping a coin until tails occurs
5. Random number generator in interval $[0, 1)$

---

2. Axioms of Probability

Before assigning probabilities:

Probability is a numerical measure of the likelihood of events,
or a function mapping events to numerical values measuring likelihood.

Axiom 1: Boundedness

For any event A:

$$0 \leqslant Pr(A) \leqslant 1$$

Axiom 2: Total Probability of Sample Space

If S is the sample space:

$$Pr(S) = 1$$

Axiom 3: Additivity for Mutually Exclusive Events

If:

$$A \cap B = \varnothing$$

then:

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

Infinite Version of Axiom 3

For mutually exclusive events:

$$A_i \cap A_j = \emptyset \quad \text{for } i \neq j$$

then:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

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3. Corollary from Probability Axioms

Define:

$$A_i = \emptyset \quad \text{for all } i > M$$

Corollary 2.1 (Finite Additivity)

For a finite number M of mutually exclusive events:

$$A_i \cap A_j = \emptyset \quad (i \neq j)$$

then:

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Notes

- Axiom 3 reduces to Corollary 2.1 when sample space is finite
- Axiom 3 is required for infinite sample spaces

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4. Propositions from Probability Axioms

Proposition 2.1: Complement Rule

For any event A:

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Proposition 2.2: Monotonicity

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